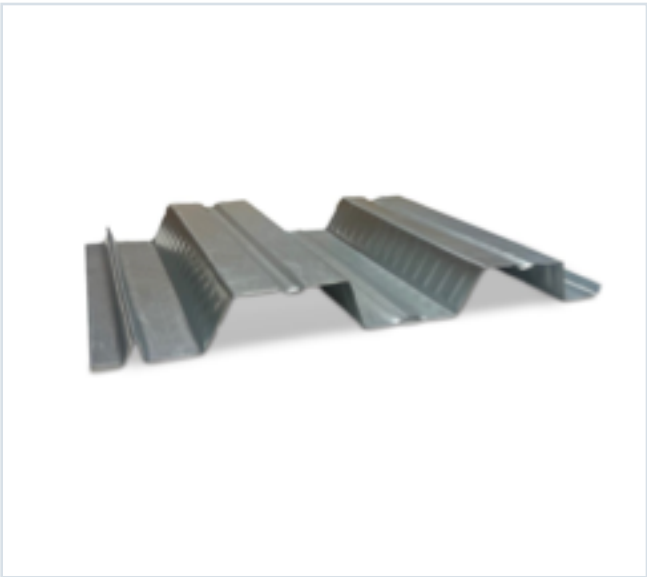


# DECKING PRODUCT SUBMITTAL

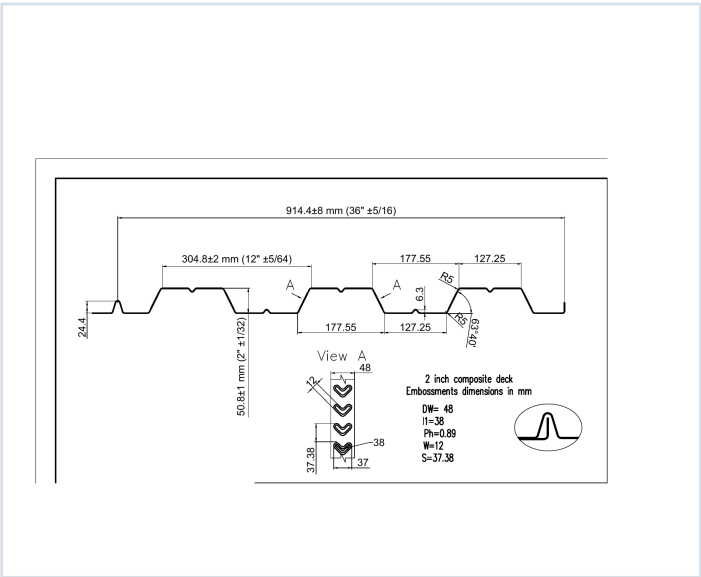
2CD-FY50-16-G90

| PRODUCT           | GRADE    | GAUGE | COATING |
|-------------------|----------|-------|---------|
| 2" Composite Deck | Grade 50 | 16 ga | G90     |

## PRODUCT VIEW



## PROFILE DRAWING



## SECTION PROPERTIES · 16 GAUGE

| Thickness<br>(in.) | Weight<br>(psf) | Fy<br>(ksi) | S+<br>(in <sup>3</sup> /ft) | S-<br>(in <sup>3</sup> /ft) | M+<br>(in-lb/ft) | M-<br>(in-lb/ft) | I+<br>(in <sup>4</sup> /ft) | I-<br>(in <sup>4</sup> /ft) |
|--------------------|-----------------|-------------|-----------------------------|-----------------------------|------------------|------------------|-----------------------------|-----------------------------|
| 0.0598             | 3.2             | 50          | 0.645                       | 0.644                       | 19321            | 19271            | 0.703                       | 0.693                       |

## SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

# PUBLISHED PERFORMANCE DATA

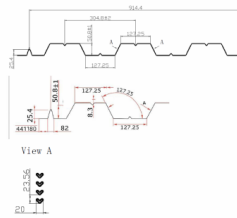
2CD-FY50-16-G90 · 16 GAUGE

## OFFICIAL STEEL CATALOG EXCERPTS

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### STEEL CATALOG PAGE 54

#### GRADE 50 STEEL



#### Section Properties

| Gage | Design Thickness (inches) | Weight (pcf) | E <sub>x</sub> (ksi) | S <sub>x</sub> (in <sup>3</sup> ) | S <sub>y</sub> (in <sup>3</sup> ) | ASD (I <sub>x</sub> + I <sub>y</sub> ) (in <sup>4</sup> ) | I <sub>x</sub> + I <sub>y</sub> (in <sup>4</sup> ) | I <sub>x</sub> + I <sub>y</sub> (in <sup>4</sup> ) |
|------|---------------------------|--------------|----------------------|-----------------------------------|-----------------------------------|-----------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| 22   | 0.0205                    | 16           | 50                   | 0.247                             | 0.254                             | 2427                                                      | 2427                                               | 0.31                                               |
| 20   | 0.0358                    | 19           | 50                   | 0.328                             | 0.336                             | 3860                                                      | 3860                                               | 0.393                                              |
| 18   | 0.0474                    | 25           | 50                   | 0.485                             | 0.500                             | 5670                                                      | 5670                                               | 0.548                                              |
| 16   | 0.0598                    | 32           | 50                   | 0.645                             | 0.664                             | 8921                                                      | 8921                                               | 0.703                                              |

Note: All section properties and ASD based strengths are calculated in accordance with AISI S100-2017, AISI S100-2012 and AISI S100-2006.

#### Shear and Web Crippling

| Gage | V <sub>u</sub> (kips) | Web Crippling (P <sub>u</sub> ) (kips) |                  | Web Crippling (P <sub>u</sub> ) (kips) |                  |
|------|-----------------------|----------------------------------------|------------------|----------------------------------------|------------------|
|      |                       | End Bearing                            | Interior Bearing | End Bearing                            | Interior Bearing |
| 22   | 388                   | 434                                    | 476              | 529                                    | 709              |
| 20   | 278                   | 295                                    | 327              | 364                                    | 508              |
| 18   | 245                   | 262                                    | 289              | 327                                    | 451              |
| 16   | 212                   | 229                                    | 256              | 290                                    | 394              |

Note: All section properties and ASD based strengths are calculated in accordance with AISI S100-2017, AISI S100-2012 and AISI S100-2006.

### STEEL CATALOG PAGE 55

| Span   | Gage | 6'  | 7'  | 8'  | 9'  | 10' | 11' | 12' | 13' | 14' | 15' | 16' |
|--------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Single | 22   | 137 | 133 | 129 | 125 | 121 | 117 | 113 | 109 | 105 | 101 | 97  |
|        | 20   | 181 | 176 | 171 | 166 | 161 | 156 | 151 | 146 | 141 | 136 | 131 |
|        | 18   | 225 | 219 | 214 | 208 | 203 | 197 | 192 | 186 | 181 | 175 | 170 |
|        | 16   | 269 | 262 | 256 | 250 | 244 | 238 | 232 | 226 | 220 | 214 | 208 |
| Double | 22   | 131 | 127 | 123 | 119 | 115 | 111 | 107 | 103 | 99  | 95  | 91  |
|        | 20   | 176 | 171 | 166 | 161 | 156 | 151 | 146 | 141 | 136 | 131 | 126 |
|        | 18   | 220 | 214 | 209 | 203 | 198 | 192 | 186 | 181 | 175 | 170 | 164 |
|        | 16   | 264 | 257 | 251 | 245 | 239 | 233 | 227 | 221 | 215 | 209 | 203 |
| Triple | 22   | 176 | 171 | 166 | 161 | 156 | 151 | 146 | 141 | 136 | 131 | 126 |
|        | 20   | 221 | 215 | 209 | 203 | 198 | 192 | 186 | 181 | 175 | 170 | 164 |
|        | 18   | 265 | 258 | 252 | 245 | 239 | 233 | 227 | 221 | 215 | 209 | 203 |
|        | 16   | 309 | 299 | 292 | 285 | 278 | 271 | 264 | 257 | 250 | 243 | 236 |

Note: 1. All section properties and ASD (I<sub>x</sub> + I<sub>y</sub>) uniform loads are calculated in accordance with AISI S100-2017, AISI S100-2012 and AISI S100-2006.  
2. Loads shown in table are uniformly distributed superimposed loads in psf. Span lengths include end-to-end spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.  
3. Bending Moment formula used for beam stress limitations are: Single and Two Span  $M = \frac{wL^2}{8}$ ; Three Span  $M = \frac{wL^2}{10}$ .

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

#### Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

| Span   | Gage | 6' | 7' | 8' | 9' | 10' | 11' | 12' | 13' | 14' | 15' | 16' |
|--------|------|----|----|----|----|-----|-----|-----|-----|-----|-----|-----|
| Single | 22   | 13 | 13 | 12 | 12 | 11  | 11  | 10  | 10  | 9   | 9   | 8   |
|        | 20   | 18 | 17 | 16 | 15 | 14  | 13  | 12  | 11  | 10  | 9   | 8   |
|        | 18   | 23 | 22 | 21 | 20 | 19  | 18  | 17  | 16  | 15  | 14  | 13  |
|        | 16   | 28 | 27 | 26 | 25 | 24  | 23  | 22  | 21  | 20  | 19  | 18  |
| Double | 22   | 13 | 13 | 12 | 12 | 11  | 11  | 10  | 10  | 9   | 9   | 8   |
|        | 20   | 18 | 17 | 16 | 15 | 14  | 13  | 12  | 11  | 10  | 9   | 8   |
|        | 18   | 23 | 22 | 21 | 20 | 19  | 18  | 17  | 16  | 15  | 14  | 13  |
|        | 16   | 28 | 27 | 26 | 25 | 24  | 23  | 22  | 21  | 20  | 19  | 18  |
| Triple | 22   | 13 | 13 | 12 | 12 | 11  | 11  | 10  | 10  | 9   | 9   | 8   |
|        | 20   | 18 | 17 | 16 | 15 | 14  | 13  | 12  | 11  | 10  | 9   | 8   |
|        | 18   | 23 | 22 | 21 | 20 | 19  | 18  | 17  | 16  | 15  | 14  | 13  |
|        | 16   | 28 | 27 | 26 | 25 | 24  | 23  | 22  | 21  | 20  | 19  | 18  |

Note: For loads that cause L/10 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

# PUBLISHED PERFORMANCE DATA

2CD-FY50-16-G90 · 16 GAUGE

## OFFICIAL STEEL CATALOG EXCERPTS

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### STEEL CATALOG PAGE 56

| Normal Weight Concrete (145 pcf) |            |         |         | Lightweight Concrete (115 pcf) |            |        |         |
|----------------------------------|------------|---------|---------|--------------------------------|------------|--------|---------|
| Maximum Unshored Clear Span      |            |         |         | Maximum Unshored Clear Span    |            |        |         |
| 400<br>(#200)<br>39 PSF          | 2x12x22 ga | 7'-8"   | 9'-1"   | 9'-3"                          | 2x12x22 ga | 8'-6"  | 9'-9"   |
|                                  | 2x12x20 ga | 9'-3"   | 10'-6"  | 10'-10"                        | 2x12x20 ga | 10'-0" | 11'-3"  |
|                                  | 2x12x18 ga | 10'-5"  | 12'-9"  | 12'-5"                         | 2x12x18 ga | 11'-2" | 13'-9"  |
|                                  | 2x12x16 ga | 11'-4"  | 14'-2"  | 13'-4"                         | 2x12x16 ga | 12'-0" | 14'-8"  |
| 450<br>(#250)<br>45 PSF          | 2x12x22 ga | 7'-7"   | 8'-8"   | 8'-10"                         | 2x12x22 ga | 8'-3"  | 9'-5"   |
|                                  | 2x12x20 ga | 8'-10"  | 10'-0"  | 10'-4"                         | 2x12x20 ga | 9'-7"  | 10'-10" |
|                                  | 2x12x18 ga | 9'-11"  | 12'-2"  | 12'-0"                         | 2x12x18 ga | 10'-9" | 13'-3"  |
|                                  | 2x12x16 ga | 10'-11" | 13'-8"  | 12'-10"                        | 2x12x16 ga | 11'-8" | 14'-6"  |
| 500<br>(#300)<br>51 PSF          | 2x12x22 ga | 7'-3"   | 8'-4"   | 8'-6"                          | 2x12x22 ga | 7'-8"  | 9'-1"   |
|                                  | 2x12x20 ga | 8'-6"   | 9'-7"   | 9'-11"                         | 2x12x20 ga | 9'-3"  | 10'-6"  |
|                                  | 2x12x18 ga | 9'-7"   | 11'-8"  | 11'-8"                         | 2x12x18 ga | 10'-5" | 12'-9"  |
|                                  | 2x12x16 ga | 10'-6"  | 13'-3"  | 12'-6"                         | 2x12x16 ga | 11'-4" | 14'-2"  |
| 550<br>(#350)<br>57 PSF          | 2x12x22 ga | 6'-11"  | 8'-0"   | 8'-2"                          | 2x12x22 ga | 7'-7"  | 8'-9"   |
|                                  | 2x12x20 ga | 8'-3"   | 9'-3"   | 9'-6"                          | 2x12x20 ga | 8'-11" | 10'-1"  |
|                                  | 2x12x18 ga | 9'-3"   | 11'-3"  | 11'-4"                         | 2x12x18 ga | 10'-0" | 12'-4"  |
|                                  | 2x12x16 ga | 10'-1"  | 12'-9"  | 12'-2"                         | 2x12x16 ga | 11'-0" | 13'-11" |
| 600<br>(#400)<br>63 PSF          | 2x12x22 ga | 6'-9"   | 7'-9"   | 7'-10"                         | 2x12x22 ga | 7'-5"  | 8'-6"   |
|                                  | 2x12x20 ga | 7'-8"   | 8'-11"  | 9'-2"                          | 2x12x20 ga | 8'-8"  | 9'-9"   |
|                                  | 2x12x18 ga | 8'-11"  | 10'-10" | 11'-0"                         | 2x12x18 ga | 9'-9"  | 11'-10" |
|                                  | 2x12x16 ga | 9'-9"   | 12'-4"  | 11'-10"                        | 2x12x16 ga | 10'-8" | 13'-6"  |
| 650<br>(#450)<br>69 PSF          | 2x12x22 ga | 6'-8"   | 7'-6"   | 7'-7"                          | 2x12x22 ga | 7'-2"  | 8'-3"   |
|                                  | 2x12x20 ga | 7'-9"   | 8'-7"   | 8'-10"                         | 2x12x20 ga | 8'-5"  | 9'-5"   |
|                                  | 2x12x18 ga | 8'-8"   | 10'-6"  | 10'-9"                         | 2x12x18 ga | 9'-5"  | 11'-6"  |
|                                  | 2x12x16 ga | 9'-6"   | 11'-8"  | 11'-7"                         | 2x12x16 ga | 10'-4" | 12'-4"  |

Note  
With crippling and clear span has not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

### STEEL CATALOG PAGE 61

| Slab Thickness (Inches) | Weight (pcf) | 7'-0" | 7'-6" | 8'-0" | 8'-6" | 9'-0" | 9'-6" | 10'-0" |
|-------------------------|--------------|-------|-------|-------|-------|-------|-------|--------|
| 4                       | 31           | 400   | 400   | 375   | 330   | 290   | 260   | 230    |
| 4.5                     | 35           | 400   | 400   | 400   | 400   | 395   | 377   | 364    |
| 5                       | 39           | 400   | 400   | 400   | 400   | 400   | 379   | 357    |
| 5.5                     | 44           | 400   | 400   | 400   | 400   | 400   | 400   | 393    |
| 6                       | 48           | 400   | 400   | 400   | 400   | 400   | 400   | 400    |
| 6.5                     | 53           | 400   | 400   | 400   | 400   | 400   | 400   | 400    |

| Slab Thickness (Inches) | 10'-6" | 11'-0" | 11'-6" | 12'-0" | 12'-6" | 13'-0" | 13'-6" | 14'-0" |
|-------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| 4                       | 290    | 280    | 272    | 257    | 243    | 230    | 220    | 210    |
| 4.5                     | 296    | 298    | 290    | 276    | 276    | 260    | 247    | 236    |
| 5                       | 304    | 295    | 290    | 277    | 268    | 259    | 245    | 232    |
| 5.5                     | 304    | 300    | 291    | 285    | 272    | 262    | 248    | 234    |
| 6                       | 307    | 307    | 304    | 304    | 278    | 255    | 234    | 216    |
| 6.5                     | 307    | 307    | 307    | 304    | 304    | 288    | 265    | 244    |

16 ga Lightweight Concrete (115 pcf, F<sub>c</sub> = 3,000 psi)

| Slab Thickness (Inches) | Weight (pcf) | 7'-0" | 7'-6" | 8'-0" | 8'-6" | 9'-0" | 9'-6" | 10'-0" |
|-------------------------|--------------|-------|-------|-------|-------|-------|-------|--------|
| 4                       | 31           | 400   | 400   | 400   | 400   | 392   | 323   | 290    |
| 4.5                     | 35           | 400   | 400   | 400   | 400   | 400   | 390   | 360    |
| 5                       | 39           | 400   | 400   | 400   | 400   | 400   | 400   | 400    |
| 5.5                     | 44           | 400   | 400   | 400   | 400   | 400   | 400   | 400    |
| 6                       | 48           | 400   | 400   | 400   | 400   | 400   | 400   | 400    |
| 6.5                     | 53           | 400   | 400   | 400   | 400   | 400   | 400   | 400    |

| Slab Thickness (Inches) | 10'-6" | 11'-0" | 11'-6" | 12'-0" | 12'-6" | 13'-0" | 13'-6" | 14'-0" |
|-------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| 4                       | 260    | 256    | 250    | 236    | 229    | 214    | 201    | 190    |
| 4.5                     | 267    | 267    | 261    | 246    | 238    | 220    | 200    | 190    |
| 5                       | 277    | 277    | 271    | 254    | 246    | 228    | 209    | 202    |
| 5.5                     | 286    | 286    | 280    | 262    | 254    | 236    | 217    | 208    |
| 6                       | 295    | 295    | 289    | 270    | 262    | 244    | 225    | 216    |
| 6.5                     | 304    | 304    | 300    | 280    | 272    | 254    | 235    | 226    |